
Flight Status of US Marketing Carriers from 2018-2022

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Which US Marketing Carriers
had the most delays and
cancellations from 2018-
2022?

Terminology

- Marketing Carrier: A carrier that sells seats using its own airline code for a flight that another carrier operates
 - Short Delay: A delay of 0-15 minutes from the expected departure time
 - Medium Delay: A delay of 15-45 minutes from the expected departure time
 - Long Delay: A delay of 45> minutes from the expected departure time
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Importance

Total Cost of Delay in the U.S. (dollars, billion)

	2016	2017	2018	2019
Airlines	5.6	6.4	7.7	8.3
Passengers	13.3	14.8	16.4	18.1
Lost Demand	1.8	2.0	2.2	2.4
Indirect	3.0	3.4	3.9	4.2
Total	23.7	26.6	30.2	33.0

Available Data

- US Bureau of Transportation Statistics
 - Marketing Carrier On-Time Performance (Beginning January 2018) data table
 - Kaggle dataset by Rob Mulla
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Existing Works

krqe.com

- Allegiant Air
- Jetblue Airways
- Frontier Airlines
- Southwest Airlines
- Spirit Airlines

timeout.com

- Allegiant Air
 - Jetblue Airways
 - Frontier Airlines
 - Envoy Air
 - United Airlines
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Methods

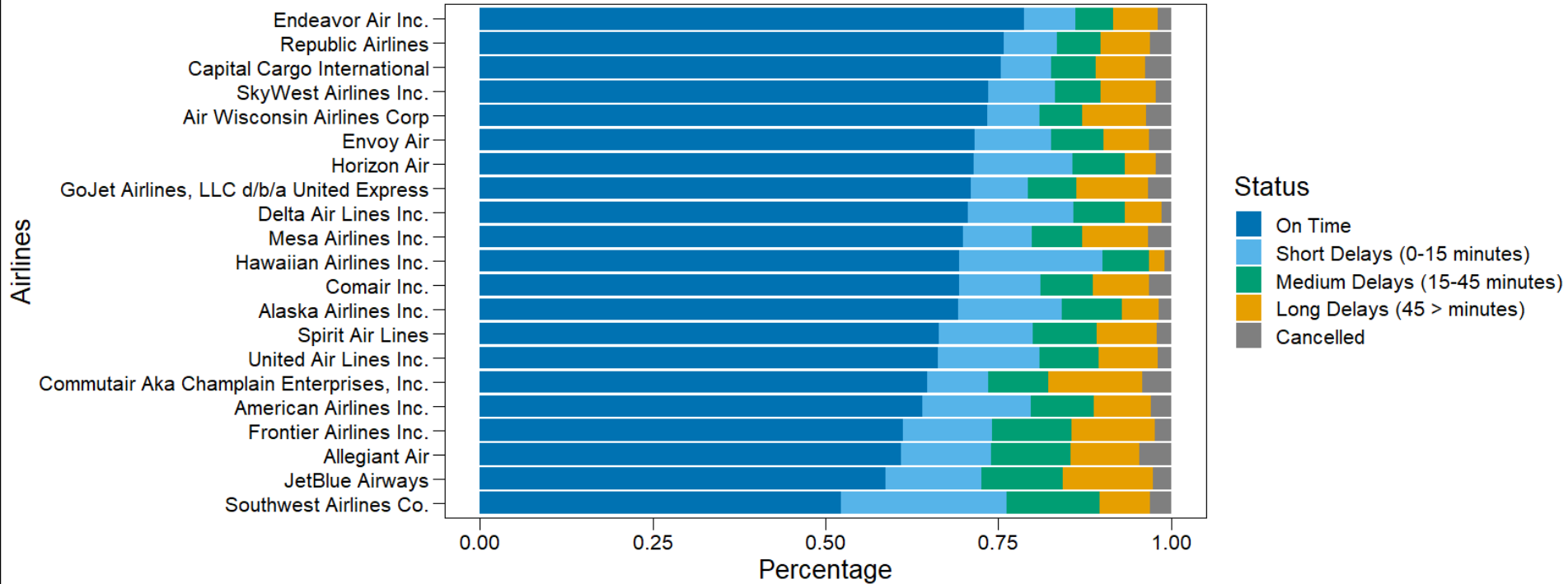
- Summary of yearly data
 - Only airlines which have 5 years of data
 - Remove missing data
 - Remove unclear data (Cancelled with delay time)
 - Find percentage of cancellations, short delays, medium delays, and long delays
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Methods

- R version 4.2.2
 - Rstudio (2022.12.0 Build 353)
 - ggplot2_3.4.0
 - tidyverse_1.3.2
 - readr_2.1.3
 - dplyr_1.1.1
 - ggthemes_4.2.4
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FINDINGS

Flight Status of US Marketing Carriers in the past 5 Years



Findings

- Southwest Airlines co (48% delayed/cancelled)
 - JetBlue Airways (41.4% delayed/cancelled)
 - Allegiant Air (39.1% delayed/cancelled)
 - Frontier Airlines inc (38.8% delayed/cancelled)
 - American Airlines (36% delayed/cancelled)
 - Commutair Inc (35.4% delayed/cancelled)
 - United Airlines (33.8% delayed/cancelled)
 - Spirit Airlines (33.6% delayed/cancelled)
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Findings

- Alaska Airlines (30.8% delayed/cancelled)
 - Comair inc (30.7% delayed/cancelled)
 - Hawaiian Airlines (30.6% delayed/cancelled)
 - Mesa Airlines inc (30.1% delayed/cancelled)
 - Delta Airlines (29.4% delayed/cancelled)
 - Gojet Airlines (29% delayed/cancelled)
 - Horizon Air (28.6% delayed/cancelled)
 - Envoy Air (28.4% delayed/cancelled)
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Findings

- Air Wisconsin Airlines (26.7% delayed/cancelled)
 - Skywest Airlines (26.4% delayed/cancelled)
 - Capital Cargo International (24.7% delayed/cancelled)
 - Republic Airlines (24.3% delayed/cancelled)
 - Endeavor Air (21.3% delayed/cancelled)
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Answers

5 Worst Airlines:

5. American Airlines inc:

- On time: 64.04%
- Short Delays: 15.67%
- Medium Delays: 9.11%
- Long Delays: 8.19%
- Cancelled: 2.97%

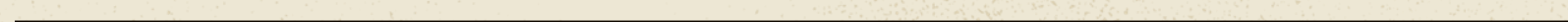


Answers

5 Worst Airlines:

4. Frontier Airlines inc:

- On time: 61.24%
- Short Delays: 12.86%
- Medium Delays: 11.53%
- Long Delays: 12%
- Cancelled: 2.35%



Answers

5 Worst Airlines:

3. Allegiant Air:

- On time: 60.91%
- Short Delays: 13.06%
- Medium Delays: 11.43%
- Long Delays: 10.01%
- Cancelled: 4.57%



Answers

5 Worst Airlines:

2. JetBlue Airways:

- On time: 58.66%
- Short Delays: 13.9%
- Medium Delays: 11.69%
- Long Delays: 13.15%
- Cancelled: 2.58%



Answers

5 Worst Airlines:

1. Southwest Airlines co:

- On time: 52.26%
- Short Delays: 23.91%
- Medium Delays: 13.48%
- Long Delays: 7.22%
- Cancelled: 3.12%



Limitations

- Covid-19 reduced the amount of flights
- Richer airlines could tackle the pandemic easier
- Only checks marketing carriers

Future Works

- How Covid-19 affected delay rate of airlines?
- Correlation between company valuation and delay rates



References

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 - <https://1000logos.net/frontier-airlines-logo/>
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 - <https://www.logo.wine/logo/JetBlue>
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 - <https://www.krqe.com/news-resources/ranking/the-airlines-with-the-most-delays/>
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THANK

YOU
